



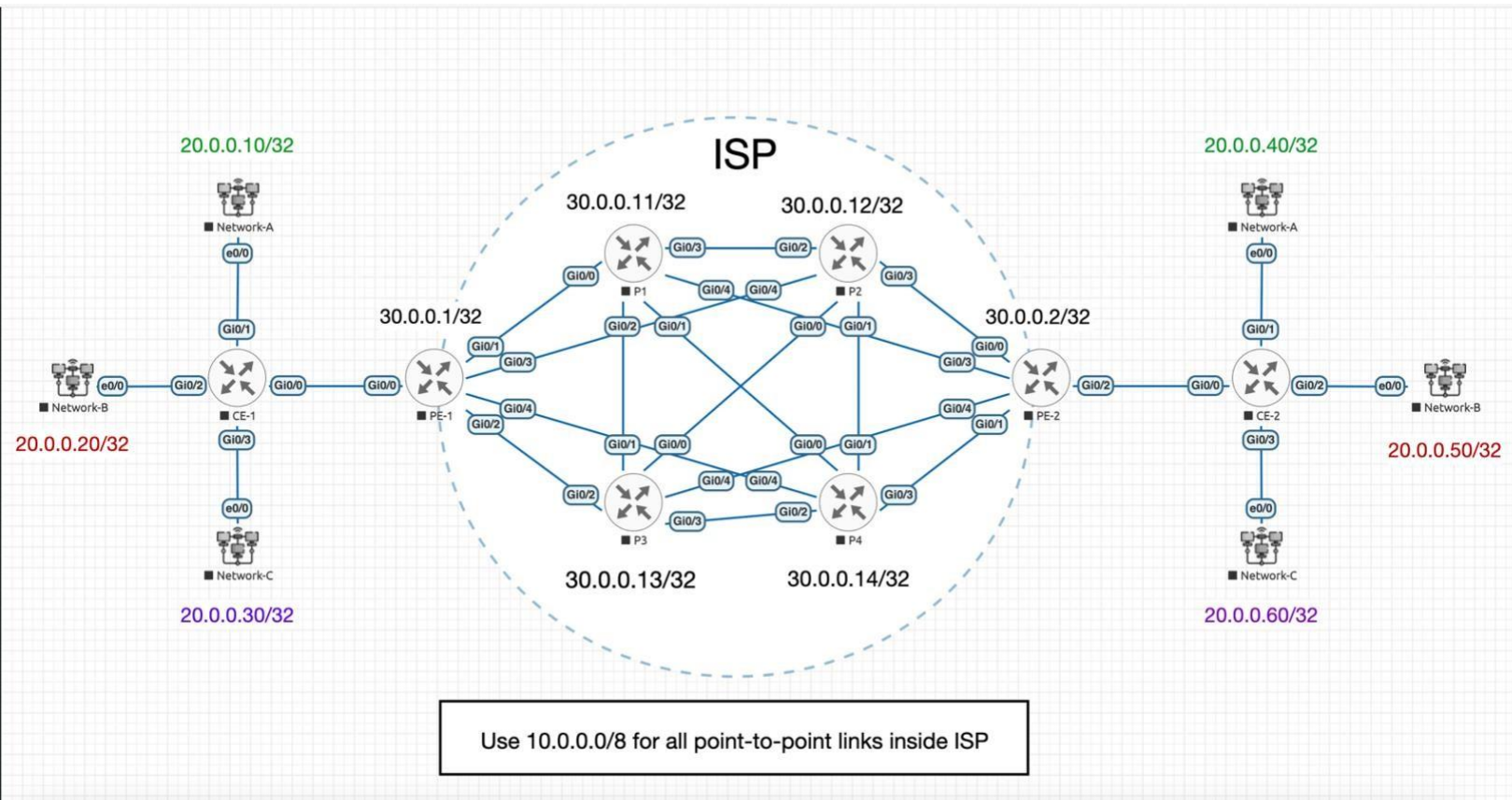
Introduction to Enterprise Networks

Spring 2025

Lab 7

Multi-Protocol Label Switching (MPLS)

Prepared by: Alireza Ta'at



Intro

As we did in the fourth lab (OSPF), GlobalTech, a multinational company, has two different sites in geographically distinct areas. However, the previous scenario had a significant challenge. GlobalTech established direct connections between each of its departments and networks in the US region to their corresponding counterparts in the Europe region. Now, imagine if this company expands in the future, increasing the number of departments to 100. Even worse, consider a scenario where there are over 10,000 companies of various sizes, each with departments spread across different geographical locations, all requiring direct connections. This would lead to an enormous and unmanageable complexity in the global network. How many IP addresses would be needed? Which organization could possibly handle such an overwhelming amount of traffic across the global network?

So, what is the solution to this problem?

To address this challenge, the role of Internet Service Providers (ISPs) becomes crucial. ISPs are responsible for controlling and managing traffic flows from specific sources to their intended destinations. They achieve this by receiving network information and routing advertisements from a source (e.g., Company A) and forwarding them efficiently to the designated destination.

However, this introduces another critical issue. As you know, the exhaustion of IPv4 addresses poses a significant limitation, and a complete transition to IPv6 is still not widely feasible. Additionally, different companies may use overlapping private IP address spaces. For instance, Company A might use the 192.168.0.0/24 subnet, and simultaneously, Company B could be using the same IP range. If both companies attempt to communicate through the ISP at the same time, how would the ISP distinguish which packet belongs to which company?

Beyond IP addressing, ISPs must also handle other network attributes such as Layer 2 VLANs, VPN labels, and VPNv4. Managing these complexities requires an efficient solution.

To solve this, a technology known as **Multi-Protocol Label Switching (MPLS)** was introduced. MPLS provides a mechanism for efficient traffic management, optimization, and separation of customer traffic within the ISP network, ensuring scalability and flexibility.

In this lab, we will focus on how ISPs use MPLS to manage traffic efficiently between the two regional sites of the GlobalTech company.

Every Network in the diagram represents a loopback adapter. Create the loopback adapters and assign IP addresses to each network according to the diagram. Also create one loopback for each router inside the ISP

*** Use 10.0.0.0/8 for the point-to-point links inside the ISP cloud and between the CE and PE routers.

Objectives

Objective 1: Setting up the topology

1. Run OSPF in the ISP cloud, advertise all loopbacks as /32. The CE routers don't speak OSPF.
2. Confirm that all loopbacks inside the ISP are reachable from one another.
3. Add a default route on each CE router to point to the adjacent PE.
4. On PE-1 add three static routes pointing to the three networks on CE-1
5. On PE-2 add three static routes pointing to the three networks on CE-2
6. Ping from 20.0.0.10 to 20.0.0.40 and from .40 to .10 and verify that OSPF uses the shortest path on both directions inside the ISP.

Paste all screenshots + commands

Objective 1.1 and 1.2:

Showing example configuration from PE-1, all OSPF configs are the same except for the IPs.

OSPF Config PE-1:

```
router ospf 1
router-id 30.0.0.1
network 10.0.0.0 0.0.0.255 area 0
network 10.0.1.0 0.0.0.255 area 0
network 10.0.2.0 0.0.0.255 area 0
network 10.0.3.0 0.0.0.255 area 0
network 30.0.0.1 0.0.0.0 area 0
!
```

Pings from PE-1 showing that it can reach ALL loopbacks:

```
PE1#ping 30.0.0.11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.11, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
PE1#ping 30.0.0.12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.12, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
PE1#ping 30.0.0.13
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.13, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
PE1#ping 30.0.0.14
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.14, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
PE1#ping 30.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
PE1#
```

Objective 1.3:

```
CE-1:
ip route 0.0.0.0 0.0.0.0 10.0.100.2
CE-2:
ip route 0.0.0.0 0.0.0.0 10.0.200.2
```

Objective 1.4:

Showing static routes to CE-1 from PE1

```
ip route 20.0.0.10 255.255.255.255 10.0.100.1
```

```
ip route 20.0.0.20 255.255.255.255 10.0.100.1
ip route 20.0.0.30 255.255.255.255 10.0.100.1
```

Objective 1.5:**Showing static routes to CE-2 from PE2**

```
ip route 20.0.0.40 255.255.255.255 10.0.200.1
ip route 20.0.0.50 255.255.255.255 10.0.200.1
ip route 20.0.0.60 255.255.255.255 10.0.200.1
```

(FOR BOTH OBJ 1.4 and 1.5: I redistributed static routes into OSPF for PE1 and PE2 OSPF)**Objective 1.6:****CE-1 to CE-2 ping:**

```
CE-1#ping 20.0.0.40 source 20.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.40, timeout is 2 seconds:
Packet sent with a source address of 20.0.0.10
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
CE-1#
```

Traceroute CE-1 to CE-2:

```
Tracing the route to 20.0.0.40
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.100.2 0 msec 0 msec 0 msec
 2 10.0.0.2 0 msec 0 msec 0 msec
 3 10.0.5.1 0 msec 0 msec 0 msec
 4 10.0.200.1 0 msec 0 msec *
CE-1#
```

CE-2 to CE-1 ping:

```
CE-2#ping 20.0.0.10 source 20.0.0.40
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.10, timeout is 2 seconds:
Packet sent with a source address of 20.0.0.40
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
CE-2#
```

Traceroute CE-2 to CE-1:

```
Tracing the route to 20.0.0.10
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.200.2 0 msec 0 msec 0 msec
 2 10.0.4.2 0 msec 0 msec 0 msec
 3 10.0.1.1 0 msec 0 msec 0 msec
 4 10.0.100.1 0 msec 0 msec *
CE-2#
```

Objective 2: Policy Based Routing (PBR)

1. Use PBR to force the traffic to pass through the following path when pinging from 20.0.0.10 to

20.0.0.40

- PE1-P1-P2-PE2

2. Use PBR to force the traffic to pass through the following path pinging from 20.0.0.40 to 20.0.0.10

- PE2-P4-P3-PE1

Paste screenshots + commands used for this objective

Objective 2.1:

PBR 20.0.0.10 to 20.0.0.40:

```
PE1:
ip access-list extended ICMP_10
permit icmp host 20.0.0.10 host 20.0.0.40
permit udp host 20.0.0.10 host 20.0.0.40 range 33434 33534
route-map PBR_10 permit 10
match ip address ICMP_10
set ip next-hop 10.0.0.2
interface gi0/0
ip policy route-map PBR_10
```

```
P1:
ip access-list extended ICMP_10
permit icmp host 20.0.0.10 host 20.0.0.40
permit udp host 20.0.0.10 host 20.0.0.40 range 33434 33534
route-map PBR_10 permit 10
match ip address ICMP_10
set ip next-hop 10.0.9.2
interface gi0/1
ip policy route-map PBR_10
```

```
P2:
ip access-list extended ICMP_10
permit icmp host 20.0.0.10 host 20.0.0.40
permit udp host 20.0.0.10 host 20.0.0.40 range 33434 33534
route-map PBR_10 permit 10
match ip address ICMP_10
set ip next-hop 10.0.6.1
interface gi0/1
ip policy route-map PBR_10
```

Traceroute:

```
Tracing the route to 20.0.0.40
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.100.2 0 msec 0 msec 0 msec
 2 10.0.0.2 0 msec 0 msec 0 msec
 3 10.0.9.2 4 msec 0 msec 0 msec
 4 10.0.6.1 4 msec 0 msec 0 msec
 5 10.0.200.1 4 msec 0 msec *
CE-1#
```

Objective 2.1:**PBR 20.0.0.40 to 20.0.0.10:**

```
PE2:
ip access-list extended ICMP_10
permit icmp host 20.0.0.40 host 20.0.0.10
permit udp host 20.0.0.40 host 20.0.0.10 range 33434 33534
route-map PBR_10 permit 10
match ip address ICMP_10
set ip next-hop 10.0.7.2
interface gi0/0
ip policy route-map PBR_10
```

```
P4:
ip access-list extended ICMP_10
permit icmp host 20.0.0.40 host 20.0.0.10
permit udp host 20.0.0.40 host 20.0.0.10 range 33434 33534
route-map PBR_10 permit 10
match ip address ICMP_10
set ip next-hop 10.0.13.1
interface fa2/1
ip policy route-map PBR_10
```

```
P3:
ip access-list extended ICMP_10
permit icmp host 20.0.0.40 host 20.0.0.10
permit udp host 20.0.0.40 host 20.0.0.10 range 33434 33534
route-map PBR_10 permit 10
match ip address ICMP_10
set ip next-hop 10.0.1.1
interface fa1/0
ip policy route-map PBR_10
```

Traceroute:

```

Tracing the route to 20.0.0.10
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.200.2 0 msec 0 msec 0 msec
 2 10.0.7.2 0 msec 0 msec 0 msec
 3 10.0.13.1 0 msec 0 msec 0 msec
 4 10.0.1.1 0 msec 0 msec 0 msec
 5 10.0.100.1 4 msec 0 msec *
CE-2#

```

Objective-3: LDP

As we mentioned, we can have full reachability between different sites using one IGP, but the problem arises when we have many prefixes to advertise, and we have a lot of networks and customers.

When a CE sends an IP packet to the PE one solution for the PE to find the other PE at the other edge through the P routers inside the ISP cloud without involving BGP is to create tunnels such as GRE between the PE routers. However, this solution requires manual configuration, security management, and maintenance. Additionally, it imposes a significant overhead on hardware resources and is generally not efficient. MPLS tunnels are a form of automatic tunnels which are optimized for packet/frame forwarding throughout the ISP cloud.

An automatic protocol called LDP or Label Distribution Protocol is invented to take on this responsibility to propagate labels to the devices of each MPLS tunnel.

Label is a part of the MPLS header created by the PE router. Do you know what the components of an MPLS header are?

1. Enable LDP for all internal routers/links except those facing the CE routers.

Because I am running OSPF in the backbone, we can just do the `mpls ldp autoconfig` command on all devices running ospf:

```

All OSPF routers:
router ospf 1
mpls ldp autoconfig

```

2. Document Labels/packets used to send data from 20.0.0.20 to 20.0.0.50 at every hop.

PE-1:

Label	Next Hop	Prefix	Outgoing Interface	Next Hop IP
30	29	20.0.0.50/32	Gi0/1	10.0.0.2
	29	20.0.0.50/32	Fa2/1	10.0.1.2
	29	20.0.0.50/32	Fa2/0	10.0.2.2

P1:

Label	Next Hop	Prefix	Outgoing Interface	Next Hop IP
29	30	20.0.0.50/32	Fa2/0	10.0.5.1

P2:

29	30	20.0.0.50/32	0	Fa1/0	10.0.6.1
----	----	--------------	---	-------	----------

P3:

29	30	20.0.0.50/32	0	Gi0/1	10.0.4.1
----	----	--------------	---	-------	----------

P4:

29	30	20.0.0.50/32	974	Fa2/1	10.0.7.1
----	----	--------------	-----	-------	----------

PE2-:

30	No Label	20.0.0.50/32	1242	Gi0/0	10.0.200.1
----	----------	--------------	------	-------	------------

So at PE-1 you can see the push of the label and then do a transfer from internal router to internal router, and then finally a pop off towards CE-2 as it reaches PE-2.

- Use the following commands to document the LDP bindings, which ports MPLS is active and the Push, Swap and Pop operations.

PE1 example of all commands below: (Above shows a every hop track of LDP)

- show mpls interfaces

```
PE1#show mpls interfaces
Interface      IP          Tunnel  BGP Static Operational
GigabitEthernet0/1  Yes (ldp)   No      No  No      Yes
Serial0/3/0        Yes (ldp)   No      No  No      Yes
FastEthernet2/0     Yes (ldp)   No      No  No      Yes
FastEthernet2/1     Yes (ldp)   No      No  No      Yes
PE1#
```

- show mpls ldp neighbor

```
PE1#show mpls ldp neighbor
PE1#show mpls ldp neighbor
Peer LDP Ident: 30.0.0.11:0; Local LDP Ident 30.0.0.1:0
TCP connection: 30.0.0.11.40426 - 30.0.0.1.646
State: Oper; Msgs sent/rcvd: 33/32; Downstream
Up time: 00:03:12
LDP discovery sources:
  GigabitEthernet0/1, Src IP addr: 10.0.0.2
Addresses bound to peer LDP Ident:
  10.0.8.1      10.0.0.2      10.0.12.1      10.0.5.2
  10.0.9.1      30.0.0.11
Peer LDP Ident: 30.0.0.13:0; Local LDP Ident 30.0.0.1:0
TCP connection: 30.0.0.13.58414 - 30.0.0.1.646
State: Oper; Msgs sent/rcvd: 33/32; Downstream
Up time: 00:03:06
LDP discovery sources:
  Serial0/3/0, Src IP addr: 10.0.3.2
Addresses bound to peer LDP Ident:
  10.0.10.1     10.0.9.2      10.0.11.1      10.0.3.2
  10.0.6.2      30.0.0.13
Peer LDP Ident: 30.0.0.12:0; Local LDP Ident 30.0.0.1:0
TCP connection: 30.0.0.12.54932 - 30.0.0.1.646
State: Oper; Msgs sent/rcvd: 33/32; Downstream
Up time: 00:02:59
LDP discovery sources:
  FastEthernet2/1, Src IP addr: 10.0.1.2
Addresses bound to peer LDP Ident:
  10.0.8.2      10.0.4.2      10.0.11.2      10.0.13.1
  10.0.1.2      30.0.0.12
Peer LDP Ident: 30.0.0.14:0; Local LDP Ident 30.0.0.1:0
TCP connection: 30.0.0.14.11910 - 30.0.0.1.646
State: Oper; Msgs sent/rcvd: 32/32; Downstream
Up time: 00:02:51
LDP discovery sources:
  FastEthernet2/0, Src IP addr: 10.0.2.2
Addresses bound to peer LDP Ident:
  10.0.10.2     10.0.13.2     10.0.12.2      10.0.2.2
  10.0.7.2      30.0.0.14
```

- **show mpls forwarding-table**

```

PE1#show mpls forwarding-table
Local   Outgoing   Prefix      Bytes Label  Outgoing   Next Hop
Label   Label      or Tunnel Id Switched      interface
16      Pop Label   10.0.4.0/24  0            Fa2/1      10.0.1.2
17      Pop Label   10.0.5.0/24  0            Gi0/1      10.0.0.2
18      20         10.0.6.0/24  0            Gi0/1      10.0.0.2
        20         10.0.6.0/24  0            Fa2/1      10.0.1.2
        21         10.0.6.0/24  0            Fa2/0      10.0.2.2
19      Pop Label   10.0.7.0/24  0            Fa2/0      10.0.2.2
20      Pop Label   10.0.8.0/24  0            Gi0/1      10.0.0.2
        Pop Label   10.0.8.0/24  0            Fa2/1      10.0.1.2
21      Pop Label   10.0.9.0/24  0            Gi0/1      10.0.0.2
22      Pop Label   10.0.10.0/24 0            Fa2/0      10.0.2.2
23      Pop Label   10.0.11.0/24 0            Fa2/1      10.0.1.2
24      Pop Label   10.0.12.0/24 0            Gi0/1      10.0.0.2
        Pop Label   10.0.12.0/24 0            Fa2/0      10.0.2.2
25      Pop Label   10.0.13.0/24 0            Fa2/1      10.0.1.2
        Pop Label   10.0.13.0/24 0            Fa2/0      10.0.2.2
26      No Label    20.0.0.10/32 0            Gi0/0      10.0.100.1
27      No Label    20.0.0.20/32 570          Gi0/0      10.0.100.1
28      No Label    20.0.0.30/32 0            Gi0/0      10.0.100.1
29      28         20.0.0.40/32 0            Gi0/1      10.0.0.2
        28         20.0.0.40/32 0            Fa2/1      10.0.1.2
Local   Outgoing   Prefix      Bytes Label  Outgoing   Next Hop
Label   Label      or Tunnel Id Switched      interface
        28         20.0.0.40/32 0            Fa2/0      10.0.2.2
30      29         20.0.0.50/32 0            Gi0/1      10.0.0.2
        29         20.0.0.50/32 0            Fa2/1      10.0.1.2
        29         20.0.0.50/32 0            Fa2/0      10.0.2.2
31      30         20.0.0.60/32 0            Gi0/1      10.0.0.2
        30         20.0.0.60/32 0            Fa2/1      10.0.1.2
        30         20.0.0.60/32 0            Fa2/0      10.0.2.2
32      32         30.0.0.2/32  0            Gi0/1      10.0.0.2
        32         30.0.0.2/32  0            Fa2/1      10.0.1.2
        32         30.0.0.2/32  0            Fa2/0      10.0.2.2
33      Pop Label   30.0.0.11/32 0            Gi0/1      10.0.0.2
34      Pop Label   30.0.0.12/32 0            Fa2/1      10.0.1.2
35      34         30.0.0.13/32 0            Gi0/1      10.0.0.2
        35         30.0.0.13/32 0            Fa2/0      10.0.2.2
36      Pop Label   30.0.0.14/32 0            Fa2/0      10.0.2.2
PE1#

```

- **traceroute [destination loopback] source [source loopback]**

```

Tracing the route to 20.0.0.50
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.100.2 0 msec 0 msec 0 msec
 2 10.0.2.2 [MPLS: Label 29 Exp 0] 0 msec 0 msec 0 msec
 3 10.0.7.1 [MPLS: Label 30 Exp 0] 0 msec 0 msec 0 msec
 4 10.0.200.1 0 msec 0 msec *
CE-1#

```

Also showing LDP is working correctly ^^^

4. Modify LDP in such a way that the traffic from 20.0.0.20 to 20.0.0.50 goes via:

- PE1-P3-P4-PE2

I used OSPF path cost to manipulate the path of the LDP labels. Every router just has an ip ospf cost 1000 for outgoing non-desirable path and ip ospf cost 10 for desirable path.

Example config of PE1:

```

interface GigabitEthernet0/1
ip ospf cost 1000
!
interface Serial0/3/0
ip ospf cost 1000

```

```
!
interface FastEthernet2/0
ip ospf cost 1000
!
interface FastEthernet2/1
ip ospf cost 10
!
```

Traceroute showing new route:

```
Tracing the route to 20.0.0.50
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.100.2 0 msec 0 msec 0 msec
 2 10.0.1.2 [MPLS: Label 29 Exp 0] 0 msec 0 msec 0 msec
 3 10.0.13.2 [MPLS: Label 29 Exp 0] 0 msec 0 msec 0 msec
 4 10.0.7.1 [MPLS: Label 30 Exp 0] 0 msec 0 msec 0 msec
 5 10.0.200.1 0 msec 0 msec *
```

Document Labels/Packets used again that shows the LDP path manipulation.

Paste screenshots + commands used

Objective 4: MPLS Traffic Engineering

MPLS-TE works based on constraint which means we use the shortest path that meets certain criteria such as, including or excluding an interface in the path, bandwidth limitation or priority.

*** Before doing this objective make sure you have studied how MPLS-TE works and the operation of the following TE components:

- Link information distribution
- Path calculation (CSPF)
- Path Setup (RSVP-TE)
- Forwarding traffic through TE tunnels

1. Create Tunnel T1 in such a way that traffic from 20.0.0.30 to 20.0.0.60 passes through the following path.
 - PE1-P4-P2-PE2

Traceroute showing the path specified:

```
Tracing the route to 20.0.0.60
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.100.2 0 msec 0 msec 0 msec
 2 10.0.2.2 [MPLS: Label 18 Exp 0] 0 msec 0 msec 0 msec
 3 10.0.10.1 [MPLS: Label 19 Exp 0] 0 msec 0 msec 0 msec
 4 10.0.6.1 0 msec 0 msec 0 msec
 5 10.0.200.1 0 msec 0 msec *
```

2. Create Tunnel T2 in such a way that traffic from 20.0.0.60 to 20.0.0.30 passes through the following path.

- PE2-P2-P3-P4-PE1

Traceroute showing the path specified:

```
Tracing the route to 20.0.0.30
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.200.2 0 msec 0 msec 0 msec
 2 10.0.6.2 [MPLS: Label 20 Exp 0] 0 msec 0 msec 4 msec
 3 10.0.11.2 [MPLS: Label 17 Exp 0] 0 msec 0 msec 0 msec
 4 10.0.13.2 [MPLS: Label 19 Exp 0] 4 msec 0 msec 0 msec
 5 10.0.2.1 4 msec 0 msec 0 msec
 6 10.0.100.1 4 msec 0 msec *
```

CE-2#

*** Both tunnels must guarantee 50% of bandwidth of the paths

T1 should have a backup path of PE1-P2-PE2 with the same performance.

T1 backup path traceroute:

Tracing the route to 20.0.0.60

VRF info: (vrf in name/id, vrf out name/id)

```
 1 10.0.100.2 0 msec 0 msec 0 msec
 2 10.0.3.2 [MPLS: Label 22 Exp 0] 0 msec 0 msec 0 msec
 3 10.0.6.1 0 msec 0 msec 0 msec
 4 10.0.200.1 0 msec 0 msec *
```

CE-1#

T2 should have a backup path of PE2-P3-PE1 with the same performance.

T2 backup path traceroute:

Tracing the route to 20.0.0.30

VRF info: (vrf in name/id, vrf out name/id)

```
 1 10.0.200.2 0 msec 0 msec 0 msec
 2 10.0.4.2 [MPLS: Label 19 Exp 0] 0 msec 0 msec 0 msec
 3 10.0.1.1 4 msec 0 msec 0 msec
 4 10.0.100.1 0 msec 0 msec *
```

3. Document Labels/Packets used on each case after tunnels are operational. Use proper traceroute commands and show the operations of Push, Pop and Swap.

This is shown above in the T1 and T2 MAIN and backup paths traceroute

4. Shut down a link on the primary path of T1 and T2 and confirm the backup paths work automatically.

This is shown above in the T1 and T2 backup path traceroute

Paste screenshots + commands

5. Document Labels/Packets, Push, Swap and Pop operations again for the backup path.

***Hint: You can use the following commands for verification:

Verification examples done all on PE-1:

- **show ip ospf mpls traffic-eng link**

```
PE1#show ip ospf mpls traffic-eng link

OSPF Router with ID (30.0.0.1) (Process ID 1)

Area 0 has 4 MPLS TE links. Area instance is 42.

Links in hash bucket 8.
Link is associated with fragment 2. Link instance is 42
Link connected to Broadcast network
Link ID : 10.0.1.1
Interface Address : 10.0.1.1
Admin Metric te: 1 igp: 1
Maximum bandwidth : 12500000
Maximum reservable bandwidth : 625000
Number of Priority : 8
Priority 0 : 625000 Priority 1 : 625000
Priority 2 : 625000 Priority 3 : 625000
Priority 4 : 625000 Priority 5 : 625000
Priority 6 : 625000 Priority 7 : 625000
Affinity Bit : 0x0
Link is associated with fragment 1. Link instance is 42
Link connected to Broadcast network
Link ID : 10.0.2.1
Interface Address : 10.0.2.1
Admin Metric te: 1 igp: 1
Maximum bandwidth : 12500000
Maximum reservable bandwidth : 625000
Number of Priority : 8
Priority 0 : 625000 Priority 1 : 625000
Priority 2 : 625000 Priority 3 : 625000
Priority 4 : 625000 Priority 5 : 625000
Priority 6 : 625000 Priority 7 : 531250
Affinity Bit : 0x0
Link is associated with fragment 4. Link instance is 42
Link connected to Broadcast network
Link ID : 10.0.0.2
Interface Address : 10.0.0.1
Admin Metric te: 1 igp: 1
Maximum bandwidth : 12500000
Maximum reservable bandwidth : 625000
Number of Priority : 8
Priority 0 : 625000 Priority 1 : 625000
Priority 2 : 625000 Priority 3 : 625000
Priority 4 : 625000 Priority 5 : 625000
Priority 6 : 625000 Priority 7 : 625000
Affinity Bit : 0x0

Links in hash bucket 11.
Link is associated with fragment 7. Link instance is 42
Link connected to Point-to-Point network
Link ID : 30.0.0.13
```

- **show mpls traffic-eng topology [loopback address]**

```
PE1#show mpls traffic-eng topology 30.0.0.1

IGP Id: 30.0.0.1, MPLS TE Id: 30.0.0.1, Router Node (ospf 1 area 0) id 12
Link[0]: Broadcast, DR: 10.0.1.1, nbr_node_id: 2, gen: 174
Frag_id 2, Intf Address: 10.0.1.1
TE metric: 1, IGP metric: 1, attribute flags: 0x0
SRLGs: None
physical_bw: 1000000 (kbps), max_reservable_bw_global: 5000 (kbps)
max_reservable_bw_sub: 0 (kbps)

Total Allocated   Global Pool   Sub Pool
BW (kbps)         BW (kbps)     BW (kbps)
-----
bw[0]:             0             5000         0
bw[1]:             0             5000         0
bw[2]:             0             5000         0
bw[3]:             0             5000         0
bw[4]:             0             5000         0
bw[5]:             0             5000         0
bw[6]:             0             5000         0
bw[7]:             0             5000         0

Link[1]: Broadcast, DR: 10.0.0.2, nbr_node_id: 1, gen: 174
Frag_id 4, Intf Address: 10.0.0.1
TE metric: 1, IGP metric: 1, attribute flags: 0x0
SRLGs: None
physical_bw: 1000000 (kbps), max_reservable_bw_global: 5000 (kbps)
max_reservable_bw_sub: 0 (kbps)
```

- **show mpls traffic-eng topology | include MPLS TE Id**
- **show mpls traffic-eng topology | include Intf**
- **show ip rsvp interface**

```
PE1#show ip rsvp interface

Terminal Services  p allocated i/f max flow max sub max VRF
Global-1          0          5M      5M      0
Se0/3/0           ena      0          5M      5M      0
Fa2/0             ena    750K          5M      5M      0
Fa2/1             ena      0          5M      5M      0
PE1#
```

- **show mpls traffic-eng tunnels summary**

```

PE1#show mpls traffic-eng tunnels summary
Signalling Summary:
  LSP Tunnels Process:      running
  Passive LSP Listener:     running
  RSVP Process:             running
  Forwarding:               enabled
  Head: 1 interfaces, 1 active signalling attempts, 1 established
        14 activations, 13 deactivations
        0 SSO recovery attempts, 0 SSO recovered
  Midpoints: 0, Tails: 1
  Periodic reoptimization:   every 3600 seconds, next in 3167 seconds
  Periodic FRR Promotion:    Not Running
  Periodic auto-bw collection: every 300 seconds, next in 167 seconds
PE1#

```

- **show ip interface brief | include Tunnel**
- **show mpls traffic-eng tunnels**

```

PE1#show mpls traffic-eng tunnels

Name: PE1_t1 (Tunnel1) Destination: 30.0.0.2
Status:
  Admin: up      Oper: up      Path: valid      Signalling: connected
  path option 1, type explicit T1_path (Basis for Setup, path weight 3)
  path option 2, type explicit T1_path_backup
  path option 3, type dynamic

Config Parameters:
  Bandwidth: 750 kbps (Global) Priority: 7 7 Affinity: 0x0/0xFFFF
  Metric Type: TE (default)
  AutoRoute: enabled LockDown: disabled Loadshare: 750 bw-based
  auto-bw: disabled
Active Path Option Parameters:
  State: explicit path option 1 is active
  BandwidthOverride: disabled LockDown: disabled Verbatim: disabled

InLabel : -
OutLabel : FastEthernet2/0, 18
RSVP Signalling Info:
  Src 30.0.0.1, Dst 30.0.0.2, Tun_Id 1, Tun_Instance 20
RSVP Path Info:
  My Address: 10.0.2.1
  Explicit Route: 10.0.2.2 10.0.10.2 10.0.10.1 10.0.6.2
                  10.0.6.1 30.0.0.2
  Record Route: NONE
  Tspec: ave rate=750 kbits, burst=1000 bytes, peak rate=750 kbits
RSVP Resv Info:
  Record Route: NONE
  Tspec: ave rate=750 kbits, burst=1000 bytes, peak rate=750 kbits
History:
  Tunnel:
    Time since created: 2 hours, 26 minutes
    Time since path change: 37 minutes, 7 seconds
    Number of LSP IDs (Tun_Instances) used: 20
  Current LSP:
    Uptime: 37 minutes, 7 seconds
  Prior LSP:
    ID: path option 2 [19]
    Removal Trigger: configuration changed

LSP Tunnel PE2_t2 is signalled, connection is up
InLabel : FastEthernet2/0, implicit-null
OutLabel : -
RSVP Signalling Info:
  Src 30.0.0.2, Dst 30.0.0.1, Tun_Id 2, Tun_Instance 2
RSVP Path Info:
  My Address: 30.0.0.1
  Explicit Route: NONE
  Record Route: NONE
  Tspec: ave rate=750 kbits, burst=1000 bytes, peak rate=750 kbits
RSVP Resv Info:

```

- **show ip route**

```

S    20.0.0.30 [1/0] via 10.0.100.1
O E2   20.0.0.40 [110/20] via 30.0.0.2, 00:28:38, Tunnel1
O E2   20.0.0.50 [110/20] via 30.0.0.2, 00:28:38, Tunnel1
O E2   20.0.0.60 [110/20] via 30.0.0.2, 00:28:38, Tunnel1
      30.0.0.0/32 is subnetted, 6 subnets
C      30.0.0.1 is directly connected, Loopback1
O      30.0.0.2 [110/3] via 30.0.0.2, 00:28:38, Tunnel1

```

PASTE SCREENSHOTS AND COMMANDS USED FOR ALL FOUR OBJECTIVES.**CONFIGURATIONS:**

```
PE-1:
mpls traffic-eng tunnels
interface Tunnel1
ip unnumbered Loopback1
tunnel mode mpls traffic-eng
tunnel destination 30.0.0.2
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 750
tunnel mpls traffic-eng path-option 1 explicit name T1_path
tunnel mpls traffic-eng path-option 2 explicit name T1_path_backup
tunnel mpls traffic-eng path-option 3 dynamic
no routing dynamic
!
*** FOR EVERY MPLS INTERFACE. ***
interface GigabitEthernet0/1
mpls traffic-eng tunnels
mpls ip
ip rsvp bandwidth 50000
!
router ospf 1
mpls traffic-eng router-id Loopback1
mpls traffic-eng area 0
router-id 30.0.0.1
ip explicit-path name T1_path enable
next-address 10.0.2.2
next-address 10.0.10.1
next-address 10.0.6.1
!
ip explicit-path name T1_path_backup enable
next-address 30.0.0.13
next-address 30.0.0.2
!

P1: - EXACT SAME FOR ALL OTHER P ROUTERS
mpls traffic-eng tunnels
!
*** FOR EVERY MPLS INTERFACE. ***
interface GigabitEthernet0/1
mpls traffic-eng tunnels
mpls ip
ip rsvp bandwidth 50000
!
router ospf 1
mpls traffic-eng router-id Loopback1
mpls traffic-eng area 0
router-id 30.0.0.11
!

PE-2:
mpls traffic-eng tunnels
```

```

!
interface Tunnel2
ip unnumbered Loopback1
tunnel mode mpls traffic-eng
tunnel destination 30.0.0.1
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 750
tunnel mpls traffic-eng path-option 1 explicit name T2_path
tunnel mpls traffic-eng path-option 2 explicit name T2_path_backup
tunnel mpls traffic-eng path-option 3 dynamic
no routing dynamic
!
*** FOR EVERY MPLS INTERFACE. ***
interface GigabitEthernet0/1
mpls traffic-eng tunnels
mpls ip
ip rsvp bandwidth 50000
!
ip explicit-path name T2_path enable
next-address 30.0.0.13
next-address 30.0.0.12
next-address 30.0.0.14
next-address 30.0.0.1
!
ip explicit-path name T2_path_backup enable
next-address 30.0.0.12
next-address 30.0.0.1
!

```

Questions:

1. Explain different components of MPLS header including Label value, EXP, S and TTL

The label value is used to make forwarding decisions and defines the LSP.

The EXP is used for QoS and help with packet prioritization

The S indicates whether the label is the last label in the stack.

TTL is similar to TTL in the IP header and prevents infinite loops in MPLS network.

2. Explain PHP or Penultimate Hop Popping and show an example of PHP in one of the objectives you have done in this lab.

PHP pops the label before it reaches a PE router to allow for the edge network to just handle IP packets.

We see this in the lab when we enable LDP, it will pop the label before it reaches the edge.

3. Explain the operation of LDP protocol. Try to understand and explain the following terms in LDP:

- What is neighbor adjacency and how LDP discovers other neighbors
 - It is similar to BGP, it is a TCP connection that initiates with a LDP hello packet with source of loopback. But its firsts use UDP to discover neighbors, then the adjacency happens.

- What ports/protocols are used by LDP before neighbor adjacency and after building adjacency for exchanging label information.
 - UDP port 646 and TCP for adjacency
- What is the role of a transport IP address in LDP
 - It builds the TCP connection between LDP neighbors, it specifies it in the LDP UDP packet to then start that adjacency
- What is LSR ID
 - Its a unique ID to identify the router, its like router-id in BGP or OSPF
- Explain LIB, LFIB and compare with RIB and FIB
 - LIB contains all labels learned no matter if they are used or not. LFIB is the chosen best path for all labels and this is the table that is used by the router. The way it differs from RIB and FIB is that LIB and LFIB use labels to determine next hop and while they use prefixes to determine next hop.

4. Explain the following terms in MPLS-TE:

- Traffic Engineering Database (TED)
 - The TED is a database that contains all data related to the TE or the tunnel that is created from MPLS-TE.
- Constrained Shortest Path First (CSPF)
 - CSPF is just an algorithm that is used to calculate the shortest path for MPLS-TE tunnels based on some attributes like bandwidth, explicit routes, priority, and admin groups.
- Path signaling, PATH and RESV messages
 - Path signaling is setting up the best path with MPLS-TE using RSVP-TE. This will allow for the headend and tailend to communicate with each other and calculate the best path. The PATH message is sent from headend to tailend. When PATH is received a RESV message will be sent in response and each RESV message received will let that router know to reserve a portion of bandwidth.
- Path setup and maintenance
 - What happens when RSVP-TE creates a path and then sends refresh messages to maintain state information.
- Path teardown
 - A PathTear message will be sent down the same RSVP-TE path telling all routers down to the tailend to free the bandwidth reservations.
- Error signaling
 - Error signaling happens usually when a router doesn't understand a parameter set by the headend so it sends it upstream with PathErr message.
- And the mechanism of Forwarding Traffic Through TE Tunnels
 - Once a MPLS-TE tunnel is set up it doesn't forward traffic, this is why we need to point the traffic through the tunnel interface with either PBR or static routes. Routing protocols can also do this as well with OSPF for example preferring shortest path tunnel interface over underlay network.